1.Explain the difference between Relative, Absolute and Mixed Cell Referencing. You have the following dataset:

Vehicle	Price
Car	500000
Bicycle	10500
Auto-Rickshaw	25000

1. Difference Between Relative, Absolute and Mixed Cell Referencing in Excel

Cell referencing in Excel is used to refer to a cell or a range of cells in a formula. There are three main types: Relative, Absolute, and Mixed Cell Referencing.

Given Dataset

Vehicle	Price (₹)
Car	5,00,000
Bicycle	10,500
Auto-Rickshaw	25,000

1. Relative Cell Referencing

A relative cell reference changes automatically when a formula is copied to another cell.

- Example: If cell B2 contains the car price (500000) and cell C2 contains the quantity (2), enter the formula `=B2*C2` in D2.
- When you copy this formula to D3, Excel automatically changes it to `=B3*C3`.

Use: It is useful when applying the same calculation to multiple rows, such as calculating the total price of different vehicles.

2. Absolute Cell Referencing

An absolute cell reference remains fixed when a formula is copied to another cell. It uses the dollar sign (`$`) before both the column letter and row number.

- Example: Suppose cell E1 contains a tax rate of 10%.

- Enter the formula `=B2*E1` to calculate the tax on the car price.
- When copied to the next row, the formula becomes `=B3*E1`. The reference `E1` remains unchanged.

Use: It is useful when referring to a fixed value, such as a tax rate, discount rate, or commission percentage.

3. Mixed Cell Referencing

A mixed cell reference fixes either the column or the row, while allowing the other part to change.

There are two types:

- `$A1` — Absolute column: Column A remains fixed, but the row can change.
- `A$1` — Absolute row: Row 1 remains fixed, but the column can change.

Example: Suppose cell B2 contains the car price and cell A2 contains a discount rate. You can use `=B2 * (1 − C1)` to calculate the discounted price.

When copied to another cell, the column B in `$B2` stays fixed, while the row in `C$1` stays fixed at row 1.

Use: It is useful when copying formulas across rows and columns, such as calculating prices at different discount rates.

Summary Table

Feature	Relative	Absolute	Mixed
Example	`A1`	`A1`	`$A1` or `A$1`
Column changes when copied	Yes	No	Depends on reference
Row changes when copied	Yes	No	Depends on reference
Dollar sign (`$`) used	No	Both row and column	Either row or column

Main use	Repeated calculations	Fixed values	Flexible row/column calculations

3.Using the data below, write a formula to calculate the average sales of items priced above 100 but less than 300
Item Price Sales Item
A 90 1000 Item
B 150 1200 Item
C 250 1500

Assuming Price is in column B and Sales is in column C, use the Excel `AVERAGEIFS` function.

Formula:

```
=AVERAGEIFS(C2:C4,B2:B4,">100",B2:B4,"<300")
```

Given Dataset

Item	Price	Sales
A	90	1000
B	150	1200
C	250	1500

Explanation:

- `C2:C4` — Range containing Sales values.
- `B2:B4,">100"` — Includes items priced above 100.
- `B2:B4,"<300"` — Includes items priced below 300.

Calculation:

Only items B and C meet both conditions.

Average Sales = (1200 + 1500) / 2 = 1350

Answer: 1350

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